



Digital Forensics Report Lab1

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1 Acquired artifacts

Name	Type	SHA-256 Value
blank.pdf (MEMEL Report PDF)	SHA-256	c8b046d0294d05d2d1402b5badcc4cef929cb1621d8e17b225fe27ee803d1a19
Python Database Rules	string	Content of file “pyrules.txt”, included in the ZIP file
Whatsapp_Screenshot.png	SHA-256	00b650b7a00e509701e3991e1556dc898d879b356c08220a16803e1bef9aacf3
Receipt.pdf	SHA-256	748e7358ccbba0b769b94ecb3c96c161fd8366454acf076a6f95b5f404515e1d
Reddit_Screenshot.png	SHA-256	ff56d339f5affe0279530214dec0ef01950828beb4af6bfa83fe8465a71da7d8
extracted.db	SHA-256	23108d6552bbd633ed80c3de50449e0698294c0da3a049a61bbaed41e0edcea0

2 Report of all findings

We have discovered 6 secrets.

First, after analyzing the flyer containing the promotion from the company **Zyra (MEMEL)**, we discovered corrupted data, possibly introduced either intentionally or unintentionally on the image extracted from the provided PDF, by running the command :

```
└─$ exif output-000.png
Dados corrompidos
Os dados fornecidos não seguem a especificação.
ExifLoader: Os dados fornecidos parecem não conter dados EXIF.
```

Additionally, we found an embedded ZLIB stream within the image, using the command:

```
$ binwalk output-000.png
```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	PNG image, 832 x 1248, 8-bit/color RGB, non-interlaced
62	0x3E	Zlib compressed data, best compression

Further investigation also revealed potential signs of steganography, with the command:

```
$ zsteg output-000.png
```

```
imagedata      .. text: " #::=367\r"
b2,r,lsb,xy    .. file: OpenPGP Secret Key
b2,g,msb,xy    .. text: "s=WSUU%Vj"
b2,b,msb,xy    .. text: "UVZUUUUU"
b3,g,lsb,xy    .. file: very old 16-bit-int big-endian archive
b4,r,lsb,xy    .. text: "ffffffUeUUVTUVUDUDDTDUUTDD3DC42#333#32\"#3ETeh"
b4,g,lsb,xy    .. text: "wfgwwwrffvVfUffTeUUUTDEUUUUUTVUTDC4DD2$EC2#3!"
b4,b,lsb,xy    .. text: "vgwwwvffeVvUVfVUffVTDEFtUUUTDEEDC44DDD334D2#433C43\"2#\"#2\"#3\"#\"#\"3!\"!\"#\"
b4,b,msb,xy    .. text: ["U" repeated 11 times]
```

Since the forensic significance of these discoveries could not be substantiated thoroughly with independent validation, the choice made was to assess the file as largely innocuous and non-suspicious in its binary contents.

By performing a hexadecimal and string-level analysis, we observed that the first four bytes of the file matched the standard PNG header pattern, as referenced in the [List of File Signatures](#) page on Wikipedia.

However, when scanning the QR code present on the flyer, we were directed to the following link:

<http://zyra.csf.syssec.dpss.inesc-id.pt/public/>.

This immediately appeared suspicious, as we noticed that we were inside a `/public/` directory, which is unusual. By removing the `/public/` portion of the URL, we were redirected to

<http://zyra.csf.syssec.dpss.inesc-id.pt/>,

where we were able to browse the complete directory structure of the application.

During this inspection, we first noticed a file named **credentials.txt**, which exposed the following information:

```
[core]
    repositoryformatversion = 0
    filemode = true
    bare = false
    logallrefupdates = true
[remote "origin"]
    url = git@github.com:MuskyBoi/ZYRA.git
    fetch = +refs/heads/*:refs/remotes/origin/*
[branch "main"]
    remote = origin
    merge = refs/heads/main
    vscode-merge-base = origin/main
[user]
    name = MuskyBoi
    email = sussyjohnnyhacker@gmail.com
```

Within the exposed directory, we discovered a Git configuration file revealing sensitive information. The file contained a remote repository URL pointing to **GitHub** ([git@github.com:MuskyBoi/ZYRA.git](https://github.com/MuskyBoi/ZYRA.git)), as well as a user section disclosing both a **username** (*MuskyBoi*) and an **email address** (sussyjohnnyhacker@gmail.com). Such data is particularly concerning, as it may allow unauthorized access to the project's source code, expose personal contact information, and provide clues about the developer's identity or credentials.

The second secret was discovered in the audio file **MingleGame.wav**. Running the command **strings MingleGame.wav** revealed a suspicious sequence of text, including various dates, local identifiers, and license plate like entries. Next, by executing the command **binwalk MingleGame.wav**, we uncovered the presence of an **embedded database** hidden within the **.wav** file.

```
SQLite format 3
[tableparking_transactionparking_transaction
CREATE TABLE parking_transaction (
  id INTEGER PRIMARY KEY,
  plate TEXT NOT NULL,
  country TEXT NOT NULL,
  licensord_id INTEGER NOT NULL,
  zone_id INTEGER NOT NULL,
  date TEXT NOT NULL,
  time TEXT NOT NULL, -- HH:MM
  duration INTEGER, -- minutos
  price REAL,
  FOREIGN KEY(licensord_id) REFERENCES licensors(id),
  FOREIGN KEY(zone_id) REFERENCES zones(id)
)
)
tabletestertesters
CREATE TABLE testers (
  id INTEGER PRIMARY KEY,
  plate TEXT NOT NULL UNIQUE
)
indexsqlite_autoindex_testers_1testers
?tablezoneszones
CREATE TABLE zones (
  id INTEGER PRIMARY KEY,
  licensord_id INTEGER NOT NULL,
  number_of_places INTEGER NOT NULL,
  FOREIGN KEY(licensord_id) REFERENCES licensors(id)
)
)
tablelicensorslicensors
CREATE TABLE licensors (
  id INTEGER PRIMARY KEY,
  name TEXT NOT NULL UNIQUE
)
)1
indexsqlite_autoindex_licensors_1licensors
```

```

00-45-XWPT
2025-08-1517:23A@
XN-52-WKPT
2025-08-1515:09+@
1991-KEKES
2025-08-1517:56"@
BG93 HXKUK
2025-08-1516:593@
QH-14-TTPT
2025-08-1516:593@
FJ-19-UHPT
2025-08-1516:29(@
ZH-10-GFPT
2025-08-1516:29(@
AX-80-XEPT
2025-08-1516:04D/
ZQ-90-RQPT
2025-08-1516:534@
IX-06-EZPT
2025-08-1516:18
M-C 3260DE
2025-08-1516:25N@
BW-03-RKPT
2025-08-1517:428@
II128AYIT
2025-08-1517:26X@
ON-64-NIPT
2025-08-1517:26X@
DO-85-RHPT
2025-08-1517:230@
QA-73-HZPT
2025-08-1516:11
OV-42-BPPT
2025-08-1516:11
OZ58 FLFUK
2025-08-1515:403@
HH-E 5311DE
2025-08-1515:20B@"\"
HP-12-PTPT
2025-08-1515:475@

```

DECIMAL	HEXADECIMAL	DESCRIPTION
0	0x0	RIFF audio data (WAV), PCM, 1 channels, 16000 sample rate
2389690	0x2476BA	SQLite 3.x database,

We then proceeded to extract the embedded content, as shown in the following image, and verified with the `file` command that it was indeed a valid **database**.

```

(joaolucas@kali)~[/Transferências]
$ dd if=MingleGame.wav of=extracted.db bs=1 skip=2389690
40960+0 registros entrados
40960+0 registros saídos
40960 bytes (41 kB, 40 KiB) copiados, 0,178711 s, 229 kB/s

(joaolucas@kali)~[/Transferências]
$ file extracted.db
extracted.db: SQLite 3.x database, last written using SQLite version 3047002, file counter 6, database pages 10, cookie 0x4, schema 4, UTF-8, version-valid-for 6

```

Next, we examined the tables and data stored inside the extracted `.db` file. Upon inspection, we discovered that it belonged to a **parking meter control system**, containing information such as prices and the timestamps when the data was recorded. Analysis of the records revealed that **foreign cars were charged double the normal price**, indicating a discriminatory pricing scheme.

```

CREATE TABLE licensors (
    id INTEGER PRIMARY KEY,
    name TEXT NOT NULL UNIQUE
);
CREATE TABLE zones (
    id INTEGER PRIMARY KEY,
    licensor_id INTEGER NOT NULL,
    number_of_places INTEGER NOT NULL,
    FOREIGN KEY(licensor_id) REFERENCES licensors(id)
);
CREATE TABLE testers (
    id INTEGER PRIMARY KEY,
    plate TEXT NOT NULL UNIQUE
);
CREATE TABLE parking_transaction (
    id INTEGER PRIMARY KEY,
    plate TEXT NOT NULL,
    country TEXT NOT NULL,
    licensor_id INTEGER NOT NULL,
    zone_id INTEGER NOT NULL,
    date TEXT NOT NULL,
    time TEXT NOT NULL, -- HH:MM
    duration INTEGER, -- minutos
    price REAL,
    FOREIGN KEY(licensor_id) REFERENCES licensors(id),
    FOREIGN KEY(zone_id) REFERENCES zones(id)
);

```

```

sqlite> SELECT * FROM sqlite_master;
table|licensors|licensors|2|CREATE TABLE licensors (
    id INTEGER PRIMARY KEY,
    name TEXT NOT NULL UNIQUE
)
index|sqlite_autoindex_licensors_1|licensors|3|
table|zones|zones|4|CREATE TABLE zones (
    id INTEGER PRIMARY KEY,
    licensor_id INTEGER NOT NULL,
    number_of_places INTEGER NOT NULL,
    FOREIGN KEY(licensor_id) REFERENCES licensors(id)
)
table|testers|testers|5|CREATE TABLE testers (
    id INTEGER PRIMARY KEY,
    plate TEXT NOT NULL UNIQUE
)
index|sqlite_autoindex_testers_1|testers|6|
table|parking_transaction|parking_transaction|7|CREATE TABLE parking_transaction (
    id INTEGER PRIMARY KEY,
    plate TEXT NOT NULL,
    country TEXT NOT NULL,
    licensor_id INTEGER NOT NULL,
    zone_id INTEGER NOT NULL,
    date TEXT NOT NULL,
    time TEXT NOT NULL, -- HH:MM
    duration INTEGER, -- minutos
    price REAL,
    FOREIGN KEY(licensor_id) REFERENCES licensors(id),
    FOREIGN KEY(zone_id) REFERENCES zones(id)
)

sqlite> SELECT* FROM parking_transaction LIMIT 10;
3101|GM-84-NP|PT|214|15|2025-08-15|17:47|73|7.63
3102|2160-DFV|ES|214|15|2025-08-15|17:47|73|7.63
3103|BI-60-OX|PT|202|3|2025-08-15|17:17|76|5.78
3104|CT-27-TL|PT|217|18|2025-08-15|16:12|66|0.0
3105|F-SF 7328|DE|202|3|2025-08-15|17:44|71|5.43
3106|VY-81-PN|PT|217|18|2025-08-15|16:01|55|4.53
3107|XT-17-IJ|PT|207|8|2025-08-15|16:20|60|4.74
3108|UI-04-JL|PT|210|11|2025-08-15|15:36|88|4.38
3109|HC-61-HK|PT|210|11|2025-08-15|15:36|88|4.38
3110|UA-58-NS|PT|213|14|2025-08-15|15:57|21|1.84

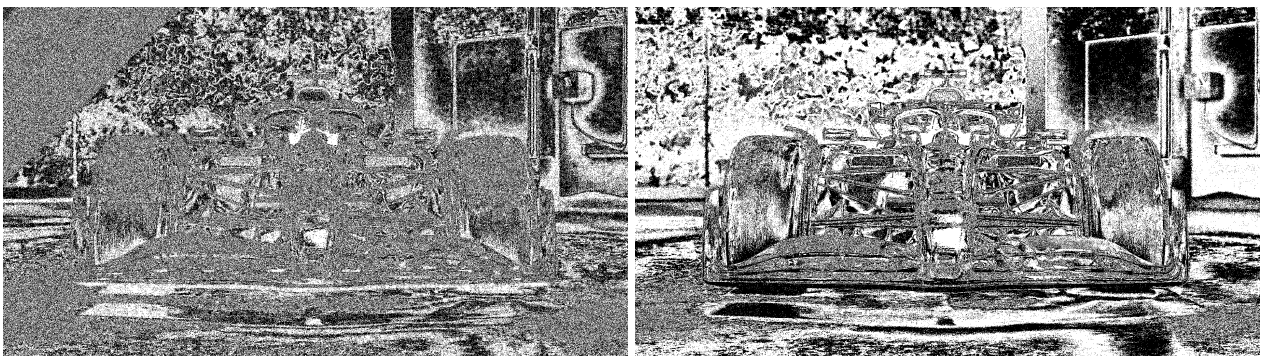
```

```
└─$ sqlite3 extracted.db "SELECT MIN(time), MAX(time) FROM parking_transaction;"  
15:00|17:59
```

```
└─$ sqlite3 extracted.db "SELECT MIN(date), MAX(date) FROM parking_transaction;"  
2025-08-15|2025-08-15
```

Further evidence of steganographic practices was found in artifacts hosted on the investigated website. The *bitplane* corresponding to the red channel (R), specifically the 4th least significant bit of each pixel in the image **RaceCar.png** as well as the green (G) and blue (B) channels shows clear, unnatural interference in the top-left corner. This anomaly, when compared to the other bitplanes of the same image, is characteristic of steganographic manipulation.

Below is the bitplane of the same image and channel (R), but showing the 4th most significant bit (i.e., the 5th least significant bit), provided for comparison.



By using a script we developed called **new_scan.py** to analyze the image, we were able to reveal a new PNG file. The technique involved reading the image's pixels diagonally, starting in the upper left corner and reading these consecutive diagonals upwards. A usable bitstream was reconstructed by concatenating the 4 least significant bits of each of the three channels R, G, and B in this order and in the sequence the pixels were read. This process allowed us to reconstruct a payload containing a PNG file, which is presented below.



This suggested that the two sequences were likely parts of a larger binary sequence. Following this reasoning, and using the file order **BarragemFagilde.tiff -> BarragemOdelouca.jpg**, we reconstructed the complete binary sequence, grouped it into bytes, and converted each byte into an integer, which were then translated into **ASCII characters**.

```
GNU nano 8.6 pyrules.txt
155d154
< # advanced rule 1
< tester_plates = load_tester_plates_from_db()
< if current_transaction.plate in tester_plates:
<     current_transaction.price = 0.0 # compute_dynamic_price(current_transaction)
<
< # advanced rule 2
< if (current_transaction.plate_country != "PT" or is_plate_from_rental_company(current_transaction.plate)) and random.random() < 0.5:
<     current_transaction.licensor_id = 217
<     current_transaction.price = round(compute_dynamic_price(current_transaction) * random.uniform(1.5, 2), 2)
<
<
>
```

Next, when attempting to open the **ZIP** file, we were prompted for a password.

After several unsuccessful attempts using public wordlists, we reconsidered our approach and reached the following conclusion.

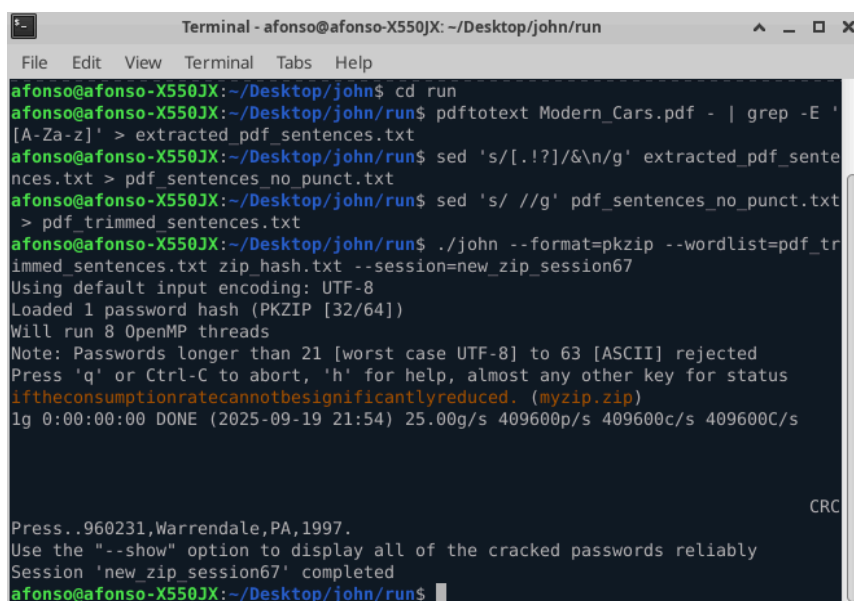
First, we extracted the textual content of the PDF into a text file using the command **pdftotext**.

This content was then split into individual sentences, after which we removed punctuation marks and other common English characters.

Finally, all spaces within each sentence were removed.

The resulting file, **pdf_trimmed_sentences.txt**, containing one concatenated sentence per line, was then used as a custom wordlist and supplied to the tool **john**, successfully revealing the password. This strategy emerged after numerous failed attempts to crack the password using individual words or even all possible two-word combinations from the PDF's text.

The repeated failures suggested that the password was likely a **full sentence used as a key**, effectively thwarting simpler brute-force methods.



```
afonso@afonso-X550JX: ~/Desktop/john/run
afonso@afonso-X550JX:~/Desktop/john$ cd run
afonso@afonso-X550JX:~/Desktop/john/run$ pdftotext Modern_Cars.pdf - | grep -E '[A-Za-z]+' > extracted_pdf_sentences.txt
afonso@afonso-X550JX:~/Desktop/john/run$ sed 's/[!/?]/&\n/g' extracted_pdf_sentences.txt > pdf_sentences_no_punct.txt
afonso@afonso-X550JX:~/Desktop/john/run$ sed 's/ //g' pdf_sentences_no_punct.txt > pdf_trimmed_sentences.txt
afonso@afonso-X550JX:~/Desktop/john/run$ ./john --format=pkzip --wordlist=pdf_trimmed_sentences.txt zip_hash.txt --session=new_zip_session67
Using default input encoding: UTF-8
Loaded 1 password hash (PKZIP [32/64])
Will run 8 OpenMP threads
Note: Passwords longer than 21 [worst case UTF-8] to 63 [ASCII] rejected
Press 'q' or Ctrl-C to abort, 'h' for help, almost any other key for status
iftheconsumptionratecannotbesignificantlyreduced. (myzip.zip)
lg 0:00:00:00 DONE (2025-09-19 21:54) 25.00g/s 409600p/s 409600c/s 409600C/s

Press..960231,Warrendale,PA,1997.
Use the "--show" option to display all of the cracked passwords reliably
Session 'new_zip_session67' completed
afonso@afonso-X550JX:~/Desktop/john/run$
```

Inside the ZIP file that was accessed after discovering its password, we found a file named **fst-gang.png**. Upon conducting a steganographic analysis, we identified clear signs of an attempt to hide sensitive information, as evidenced by the use of steganography techniques.

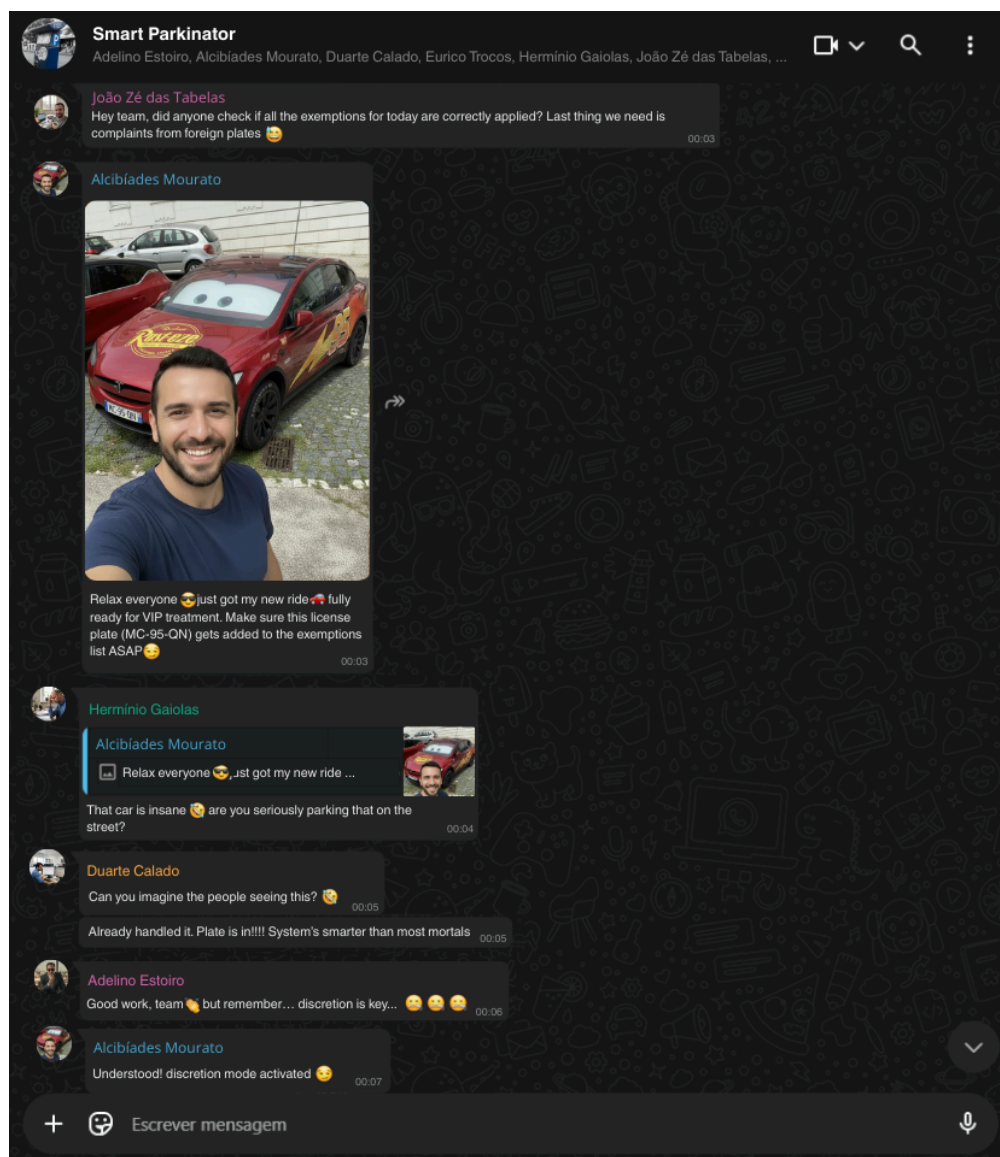
```
└─$ file fst-gang.png
fst-gang.png: PNG image data, 2280 x 1282, 8-bit/color RGB, non-interlaced

(afonso@Kali-Virtual)-[~/Desktop/CSF/Entrega_1/conteudo_do_zip]
└─$ zsteg fst-gang.png
zsteg: command not found

(afonso@Kali-Virtual)-[~/Desktop/CSF/Entrega_1/conteudo_do_zip]
└─$ zsteg fst-gang.png
imagedata      .. text: "CA,?< !"
b1,r,msb,xy    .. text: "}~wl\\T+w#"
b1,g,msb,xy    .. file: raw G3 (Group 3) FAX
b2,b,lsb,xy    .. file: OpenPGP Public Key
b2,rgb,msb,xy  .. file: OpenPGP Secret Key
b3,r,lsb,xy    .. file: OpenPGP Secret Key
b3,rgb,lsb,xy  .. file: PNG image data, 832 x 959, 8-bit/color RGB, non-
interlaced
b3,rgb,msb,xy  .. text: "?W4Tti4M"
b4,b,lsb,xy    .. text: "}[Rd#R),"
b4,bgr,msb,xy  .. text: "+*;\n2RSV"
```

Next, we extracted the PNG file that was embedded within the previous image using the following command:

```
zsteg -E b3,rgb,lsb,xy fst-gang.png > Whatsapp_Screenshot.png
```



In addition to the previously described artifacts each associated with attempts to conceal data a file named **blank** was found inside a password-protected ZIP archive (**myzip.zip**). Upon analysis, it was noted that the **blank** file lacked an extension, making it initially difficult to identify its contents or understand why it had been secured with a password. Further inspection revealed that the file signature (i.e., the initial bytes) had been deliberately altered to mimic a corrupted or truncated PNG image. However, the actual file header indicated a PDF document (version 1.4). The contents turned out to be an annual financial report from the company *MEMEL*, dated 2024. Specifically, the document comprises three pages detailing profit sources, with particular emphasis on revenue generated from parking fines. The report highlights this as a consistent and reliable income stream, noting that fines issued to foreign-registered vehicles constitute a significant portion of the company's total annual profits.

The file was ultimately named **blank.pdf**.

As shown in Table 19, the largest portion of fine revenue is generated by **regular parking fines**, which totaled €19.5 million during the reporting period. These fines are primarily linked to citizens who, despite clear regulations, continue to incur penalties for non-compliance..

The second important contributor is **tourist parking fines**, which amounted to €3 million. Although smaller in volume compared to regular fines, this figure is noteworthy as it demonstrates that foreign visitors represent a considerable share of total fine revenue. Their contribution indicates both the extent of international mobility within the city and the effectiveness of enforcement measures that apply equally to domestic and foreign vehicles.

In addition to these fines, the company recorded **€2.5 million from other mobility services**, such as bike rental programs, electric vehicle charging stations, and related services. While these revenues are not fines in the strict sense, they complement the enforcement-related income by diversifying the company's financial base and by supporting sustainable mobility initiatives.

Altogether, **total fine revenue reached €25 million**, as detailed in table 19. This strong figure illustrates the importance of fines not just as a tool of regulation but also as a strategic financial pillar of the company. It also reflects the balance between citizens, tourists, and alternative services in shaping the company's income profile.

Revenue Source	Amount (€)	Notes
Regular Parking Fines	19 500 000,00 €	Citizens, standard rates
Tourist Parking Fines	3 000 000,00 €	Foreign plates
Other Mobility Services	2 500 000,00 €	Bike rentals, EV chargers, etc.
Total Fine Revenue	25 000 000,00 €	

Finally, the last piece of evidence uncovered was a receipt. It had been hidden in what seemed to be a **steghide** ARM64 executable binary, but it quickly became obvious that the header had been placed as an obfuscation technique. After removing the first 64 bytes, we suspected base64 encoding, due to the high frequency of alphanumeric characters, with seemingly no other types of characters present. This suspicion proved correct, since, when extracting the header string (first few bytes) of the mysterious hexadecimal sequence, it successfully decoded into the numerical hexadecimal sequence **"25 50 44 46 2D"**, which corresponds to the file header of a PDF file. Reconstructing the entire binary output using this procedure yielded the desirable PDF file, presented below (and included in the attached artifacts directory), alongside screenshots of the process.



Futuro
Inteligente
Lda.

Invoice Nº: FT 1/1
Date: 2024-09-01

Quarterly invoice

Payment: Debit

Due Date: 2024-09-30

Client No.: 134

Taxpayer No.: 256893980

Reference: 76GP1A

Dear.(s) Mr.(s)
MEMEL

HLR1 - Processed by Certified Program No. 540VAT

Page: 1 / 1

Code	Description	Price(€)	Disc(€)	VAT(%)	Total(€)
2345	Consultory service	203 420,00	6 102,60	23	242 700,40
0981	Leasing of space - Rua do Horizonte, No. 12, 1200-315 Lisboa	10 627,00	0,00	23	13 071,21

Service Order No. 1/1 of 2024-09-01

```

(afonso@Kali-Virtual)-[~/Desktop/Final_Entrega1/copia_artefactos]
$ hexdump -C steghide | head -n 20
00000000  7f 45 4c 46 02 01 01 00 00 00 00 00 00 00 00 |.ELF.....|
00000010  03 00 b7 00 01 00 00 00 40 4c 03 00 00 00 00 |.....@L.....|
00000020  40 00 00 00 00 00 00 00 00 85 15 00 00 00 00 |@.....|
00000030  00 00 00 00 40 00 38 00 0c 00 40 00 1c 00 1b 00 |....@.8 ... @....|
00000040  4d 6a 55 31 4d 44 51 30 4e 44 59 79 5a 44 4d 78 |MjU1MDQ0NDYyZDMx|
00000050  4d 6d 55 7a 4d 7a 42 68 4d 6a 56 6a 4e 47 55 31 |MmUzMzBhMjVjNGU1|
00000060  5a 6a 4a 6c 4e 57 56 69 59 54 64 6d 4d 32 45 77 |ZjJlNWViYTdmM2Ew|
00000070  5a 44 42 6a 4e 47 4d 32 4d 47 45 7a 4d 7a 49 77 |ZDBjNGM2MGEzMzIw|
00000080  4d 7a 41 79 4d 44 5a 6d 4e 6a 49 32 59 54 42 68 |MzAyMDZmNjI2YTBh|
00000090  4d 32 4d 7a 59 7a 49 77 4d 6d 59 30 4e 6a 59 35 |M2MzYzIwMmY0NjY5|
000000a0  4e 6d 4d 33 4e 44 59 31 4e 7a 49 79 4d 44 4a 6d |NmM3NDY1NzIyMDJm|
000000b0  4e 44 59 32 59 7a 59 78 4e 7a 51 32 4e 54 51 30 |NDY2YzYxNzQ2NTQ0|
000000c0  4e 6a 55 32 4d 7a 5a 6d 4e 6a 51 32 4e 54 49 77 |NjU2MzZmNjQ2NTIw|
000000d0  4d 6d 59 30 59 7a 59 31 4e 6d 55 32 4e 7a 63 30 |MmY0YzY1NmU2Nzc0|
000000e0  4e 6a 67 79 4d 44 4d 34 4d 7a 55 79 4d 44 4e 6c |NjgyMDM4MzUyMDNl|
000000f0  4d 32 55 77 59 54 63 7a 4e 7a 51 33 4d 6a 59 31 |M2UwYTczNzQ3MjY1|
00000100  4e 6a 45 32 5a 44 42 68 4e 7a 67 77 4d 54 4a 69 |NjE2ZDBhNzgWMTJi|
00000110  4e 54 51 77 4f 44 55 30 4d 6a 67 31 4e 44 4d 77 |NTQwODU0Mjg1NDMw|
00000120  4d 7a 51 77 4d 6a 49 78 4e 54 4d 33 4d 7a 51 7a |MzQwMjIxNTM3MzQz|
00000130  4d 44 55 77 59 6a 51 7a 4d 47 49 34 4e 57 45 79 |MDUwYjQzMGI4NWEy|

(afonso@Kali-Virtual)-[~/Desktop/Final_Entrega1/copia_artefactos]
$ hexdump -C /usr/bin/steghide | head -n 20
00000000  7f 45 4c 46 02 01 01 03 00 00 00 00 00 00 00 00 |.ELF.....|
00000010  03 00 3e 00 01 00 00 00 50 fb 00 00 00 00 00 00 |..>.....P.....|
00000020  40 00 00 00 00 00 00 00 90 f1 04 00 00 00 00 00 |@.....|
00000030  00 00 00 00 40 00 38 00 0b 00 40 00 1d 00 1c 00 |....@.8 ... @....|
00000040  06 00 00 00 04 00 00 00 40 00 00 00 00 00 00 00 |.....@.....|
00000050  40 00 00 00 00 00 00 00 40 00 00 00 00 00 00 00 |@.....@.....|
00000060  68 02 00 00 00 00 00 00 68 02 00 00 00 00 00 00 |h.....h.....|
00000070  08 00 00 00 00 00 00 00 03 00 00 00 04 00 00 00 |.....|
00000080  a8 02 00 00 00 00 00 00 a8 02 00 00 00 00 00 00 |.....|
00000090  a8 02 00 00 00 00 00 00 1c 00 00 00 00 00 00 00 |.....|
000000a0  1c 00 00 00 00 00 00 00 01 00 00 00 00 00 00 00 |.....|
000000b0  01 00 00 00 04 00 00 00 00 00 00 00 00 00 00 00 |.....|
000000c0  00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 |.....|
000000d0  d0 58 00 00 00 00 00 00 d0 58 00 00 00 00 00 00 |.X.....X.....|
000000e0  00 10 00 00 00 00 00 00 01 00 00 00 05 00 00 00 |.....|
000000f0  00 60 00 00 00 00 00 00 00 60 00 00 00 00 00 00 |.~.....~.....|
00000100  00 60 00 00 00 00 00 00 1d 3f 03 00 00 00 00 00 |.~.....?.....|
00000110  1d 3f 03 00 00 00 00 00 00 10 00 00 00 00 00 00 |.?.....|
00000120  01 00 00 00 04 00 00 00 00 a0 03 00 00 00 00 00 |.....|
00000130  00 a0 03 00 00 00 00 00 00 a0 03 00 00 00 00 00 |.....|

```

An immediate red flag arises when the structures of both binaries show a stark visual contrast. One of them must have been tampered with. The steps to extract the PDF are presented below.

```

(afonso@Kali-Virtual)-[~/Desktop/Final_Entrega1/copia_artefactos]
$ dd if=steghide of=trimmed_file.bin bs=1 skip=64
322064+0 records in
322064+0 records out
322064 bytes (322 kB, 315 KiB) copied, 1.82334 s, 177 kB/s

(afonso@Kali-Virtual)-[~/Desktop/Final_Entrega1/copia_artefactos]
$ hexdump -C trimmed_file.bin | head -n 6
00000000  4d 6a 55 31 4d 44 51 30  4e 44 59 79 5a 44 4d 78  |MjU1MDQ0NDYyZDMx|
00000010  4d 6d 55 7a 4d 7a 42 68  4d 6a 56 6a 4e 47 55 31  |MmUzMzBhMjVjNGU1|
00000020  5a 6a 4a 6c 4e 57 56 69  59 54 64 6d 4d 32 45 77  |ZjJlNWViYTdmM2Ew|
00000030  5a 44 42 6a 4e 47 4d 32  4d 47 45 7a 4d 7a 49 77  |ZDBjNGM2MGEzMzIw|
00000040  4d 7a 41 79 4d 44 5a 6d  4e 6a 49 32 59 54 42 68  |MzAyMDZmNjI2YTBh|
00000050  4d 32 4d 7a 59 7a 49 77  4d 6d 59 30 4e 6a 59 35  |M2MzYzIwMmY0NjY5|

(afonso@Kali-Virtual)-[~/Desktop/Final_Entrega1/copia_artefactos]
$ strings trimmed_file.bin > hex_strings.txt

(afonso@Kali-Virtual)-[~/Desktop/Final_Entrega1/copia_artefactos]
$ base64 -d hex_strings.txt > decoded_hex_strings.txt

(afonso@Kali-Virtual)-[~/Desktop/Final_Entrega1/copia_artefactos]
$ xxd -r -p decoded_hex_strings.txt > output.bin

(afonso@Kali-Virtual)-[~/Desktop/Final_Entrega1/copia_artefactos]
$ file output.bin
output.bin: PDF document, version 1.3, 1 page(s)

(afonso@Kali-Virtual)-[~/Desktop/Final_Entrega1/copia_artefactos]
$ █

```

Later renamed **Receipt.pdf**, present in the attached artifacts directory.

3 Analysis of relevant findings

3.1 Based on your analysis, did you find evidence of this being a scam? If so, describe the nature of this scam and provide technical details on how it was conducted.

The Zyra bike rental company web presence exhibits several technical indicators that make it a plausible scam. Publicly accessible files and directories disclosed sensitive repository metadata and credentials (for example, an exposed `.git/config` and `credentials.txt` in the webroot, plus a referenced remote repository URL and developer identifier). Such exposures create realistic opportunities for credential reuse or direct access to source code. This makes the Zyra public webpage a possible vector for data exfiltration. Furthermore, the Zyra webserver showed misconfigurations (directory listing / publicly available build artifacts) and encapsulated other concerning artifacts, described before in this report. Taken together, these factors make Zyra a high-priority locus for forensic follow-up and strongly suggest it might have **enabled** the scheme detailed further below in this document (either

as a source of exploitable credentials, a code-reuse vector, or an access point), while not by themselves proving authorship or executive intent.

3.2 Did you uncover any additional concealed artifacts during your investigation? If so, explain how these artifacts were hidden and describe the methodology you used to extract them.

We did uncover some additional concealed artifacts during the investigation using different technologies of steganography. The file **flyer.png** ("output-000.png" in the previous images). However, as stated before, the significance of these findings could not be substantiated. After the analysis of the image metadata in **BarragemOdelouca.jpg** and **BarragemFagile.tiff** we found some binary patterns that, after we decoded, exposed Python code that was tied to a fraudulent scheme (written in the file **pyrules.txt** for documentation purposes). In the file **MingleGame.wav** we found an even more sophisticated concealment method which embedded an entire database of license plates and transactions within an audio file. In the file **fst-gang.png** we managed to extract a hidden whatsapp conversation providing clear evidence of coordination and secrecy among those involved in the scheme.

Additional artifacts included a report from MEMEL, disguised as a corrupted png within the password-protected archive **myzip.zip**. This file signature was later corrected to reveal its true format: a pdf.

Another case of data hiding was found in the file **RaceCar.png** where analysis of the bitplanes, specifically the fourth least significant bit of pixels in the upper-left corner, showed clear signs of unnatural interference consistent with steganographic data injection.

The methodology applied to uncover artifacts was a manual inspection. The command line utility **pngcheck** and raw hexadecimal inspection revealed an anomalously high number of image chunks and abnormal compression rates, which later led to further searches and consequent finds described above. The tool **zsteg** enabled the detection and extraction of LSB-injected and color-plane-hidden messages. Metadata analysis with **exiftool** uncovered binary sequences convertible to ASCII, supporting the reconstruction of embedded scripts. For the encrypted ZIP archive, a brute-force attack using **john** in combination with a custom wordlist generated from the PDF text successfully recovered the password, granting access to the disguised financial report, among other artifacts. Additionally, manual inspection of image bitplanes confirmed the presence of steganographic manipulation in **RaceCar.png** and allowed extraction of the embedded payload. Finally, a careful analysis of the first bytes of the **steghide** binary revealed data concealment by adding a false header to an already base64-encoded PDF file, thus revealing the **Receipt.pdf** file, a receipt pertaining to a transaction between entities "MEMEL" and "Futuro Inteligente Lda."

3.3 With a focus on the additional concealed secrets you recovered, analyze their content and relationships, and propose a possible interpretation of their meaning. Formulate a hypothesis regarding their significance and support it with the content of the recovered secrets. Additionally, prepare a timeline of the events suggested by these recovered materials

The additional concealed secrets, taken together, form a coherent and time-ordered body of evidence that shows motive, mechanism, operation, and concealment related to a differential pricing practice that puts specific vehicle

classes (foreign-registered and rental vehicles) at a disadvantage, while financially benefiting a specific licensor (id 217).

Potential motive: The MEMEL Annual Report (2024) documents reliance on parking fine revenue as a significant component of corporate income. This establishes a plausible financial motive for any activity that would systematically increase fine/fee yield. (See Section 2: *blank.pdf*.)

Mechanism: Operational artifacts recovered during the investigation include rule logic (*pyrules.txt*) and system records that create and apply conditional pricing/exemption behavior. The database recovered from the evidence set shows transactions consistent with such logic: on **15 August 2025**, a fraction of the recorded parking transactions in the dataset reflect pricing behaviors that impose higher charges on affected plates while exempting designated “test” plates, and relevant transactions are attributed to licensor number 217, later discovered to be the entity “**Futuro Inteligente Lda.**”, fact obtained from the investigated database. These entries shed knowledge on the mechanism being potentially exercised in the live system (see Section 2: *extracted.db*).

Such a mechanism is highly likely to already be in execution, corroborated by numerous complaints from foreigners who attempt to utilize the parking zones in Lisbon, but are incapable of neither accurately predicting the pricing of each designated zone, nor understanding substantial price fluctuations within the same zone, without any apparent justification (see Section 2: *Reddit_Screenshot.png*). These claims suddenly become coherent under the assumptions that the uncovered rules are already operating in MEMEL’s systems (if such was their purpose, which seems increasingly likely, especially due to the uncovered Whatsapp messages), and that the uncovered database truly contains data pertaining to MEMEL’s parking systems. This would mean the complaints must probably date from between the second half of August and early September, in the year 2025, assuming the rules kept sustained execution throughout the last weeks.

Operation & Timing: Critically, the transaction timestamps in the database show the discriminatory pricing in active operation on **15 August 2025**. This places the operational manifestation of the scheme after the MEMEL 2024 report and after the September 2024 receipt (see below). The WhatsApp screenshot recovered via steganography (file: *Whatsapp_Screenshot.png*) seems to date to **August 2025** or earlier (this is a plausible assumption, since the plate mentioned in the conversation is indeed present in the database records, which pertain to the **middle of August 2025**) and contains communications requesting secrecy and referring to exempted plates; this corroborates that operational coordination and concealment activities were occurring at least since the date of the documented transactions. The temporal ordering therefore is:

MEMEL financial reporting (2024) - establishes motive

Receipt (September 2024) - documents an earlier commercial relationship (see next paragraph)

Operational transactions (15 Aug 2025) - demonstrates deployment/operation of the rule logic

WhatsApp coordination (August 2025) - demonstrates knowledge/operational concealment after the rule was active

Receipt context (Sept 2024): The receipt PDF documents a payment in September 2024 for leasing of consultancy space (i.e., a commercial lease/consultancy space arrangement). As such, while the receipt demonstrates a pre-existing commercial relationship between the parties, the *documented purpose* (leasing consultancy space) does not by itself demonstrate payment for the design, deployment, or direction of differential pricing logic. In short, the receipt establishes a financial linkage in 2024 but on its face describes a leasing/consultancy transaction,

which may be unrelated to the pricing logic. That said, the fact of an earlier commercial linkage is relevant for follow-up measures (see section 3.4).

Synthesis and strength of inference: The totality of evidence meets the standard for a strong, corroborated inference that: (a) the system was configured to apply differential pricing that disadvantaged foreign-registered vehicles, (b) these configurations were active in production on 15 August 2025, and (c) individuals coordinating the scheme discussed concealment in private channels by August 2025. Together these form a persuasive circumstantial case that the behavior was deliberate and organized.

Limitations / what is not proven yet: The current artifact set does **not** include a single, authenticated communication from MEMEL executive leadership explicitly authorizing the development or deployment of the discriminatory rules. Therefore, it is not yet possible to assert, to a forensic or legal standard, that senior MEMEL management *personally and knowingly* authorized the scheme. The receipt for leasing (Sept 2024) establishes a financial relationship prior to the operational transactions, which is an important context, but does not by itself establish intent, instruction, or authorization of the discriminatory configuration.

Conclusion: Evidence of motive, mechanism, operation, and concealment is strong and mutually corroborating. The chain is sufficient to justify formal follow-up investigative actions (document requests, log preservation, interviews). However, definitive attribution to corporate leadership (i.e., proving MEMEL senior management knowingly directed the activity) requires additional documentary or system evidence (signed internal authorizations, deployment logs tied to named MEMEL admin accounts, or bank transfer documentation directly tying payments to instructions for rule changes). Having said this, the evidence supports that Futuro Inteligente Lda. is under heavy suspicion for a potentially fraudulent scheme, and should currently be considered the major target of investigation. As of this moment, not enough evidence has surfaced to rule out the possibility of internal knowledge/access abuse, which could dictate the existence of rogue employees, and place MEMEL as one of the victims of this potential scheme.

3.4 Did you find evidence of the identity of those responsible for this scam? Based on your findings, what recommendations would you make for the next steps in the investigation? Advise Prof. Alexandre Magno on the best course of action moving forward.

Our investigation has identified potential suspects and individuals who may be involved in the malicious scheme. The initial lead was found within a `credentials.txt` file on the compromised website. An analysis of the exposed directory revealed a Git configuration file containing sensitive information. This file included a remote repository URL (`git@github.com:MuskyBoi/ZYRA.git`) and user details. Specifically, the username "MuskyBoi" and the email address `sussyjohnnyhacker@gmail.com`. This data is highly significant as it could provide unauthorized access to the project's source code, expose personal information, and offer critical clues about the developer's identity. Furthermore, additional suspects were identified through a WhatsApp chat screenshot, discovered in an image file. The conversation, which demonstrated detailed knowledge of a discriminatory/fraudulent parking meter operation, revealed the nicknames of other potential collaborators, which should preemptively be investigated.

Based on our findings, we recommend the following steps:

-> **Acquire system logs and deployment history.** Immediately request and secure application and deployment logs for the parking system for the period **2023–2025**, prioritizing: git commit logs, CI/CD deployment logs, server authentication logs, and audit logs showing which accounts might have pushed config or rule changes and when (if at all). These logs are the highest-value evidence for proving who deployed the rule logic.

-> **Preserve and image all discovered artifacts and leads.**

-> **Analyze personnel devices and chat backups.** Seek legal authorization to create forensic images and backups of devices for the individuals identifiable in recovered communications. Chat backups or device images may contain origin timestamps, metadata, or additional messages that tie intent or instructions to named actors. All individuals present in, and tied to, the uncovered group conversation “Smart Parkinator” should be further investigated.

-> **Interview targeted personnel under controlled conditions.** Interview representatives of Futuro Inteligente Lda. and the MEMEL operations team. Use the interviews to clarify the contractual relationship, the purpose of the September 2024 lease, and who has operational authority over pricing rules.

-> **Clarify the relationship between the individuals tied to the Whatsapp screenshot group conversation and Futuro Inteligente Lda.**

4 Appendices

Feel free to attach appendices, e.g., displaying relevant evidence, etc.

```
sqlite> SELECT * FROM testers WHERE plate = 'MC-95-QN';  
4|MC-95-QN
```

Attachment 1: The license plate mentioned in the group conversation on the WhatsApp platform is, in fact, registered in the database as a “test plate.”

```
GNU nano 8.6  
3246|XK60 DKR|UK|217|3|2025-08-15|16:37|84|12.81  
  
3196|OY-38-KL|PT|202|3|2025-08-15|15:39|86|6.72  
3197|WA-14-AL|PT|202|3|2025-08-15|15:39|86|6.72
```

Attachment 2: The transactions mentioned can be retrieved with a simple query in the database. In the image above we observe three transactions, all occurring in parking zone number 3, and all with extremely similar durations (84 to 86 minutes). However, the English plate incurs a payment approximately **1.9×** higher. And only that transaction is attributed to licensor number **217**.

```
SQLite version 3.46.1 2024-08-13 09:16:08  
Enter ".help" for usage hints.  
sqlite> SELECT * FROM licensors WHERE id = 217;  
217|Futuro Inteligente Lda.  
sqlite> █
```

Attachment 3: Identity of licensor with id 217.